KNOWLEDGE BASE

QUANTUM CHEMISTRY: FROM WAVES IN A BOX TO DFT LABELS

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Standing waves, hydrogen, molecules, basis sets, Hartree–Fock, density functional theory and coupled cluster

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1 RECOMMENDED READING AND SUPPORTIVE TEXTS

Recommended reading

The Feynman Lectures on Physics, Vol. III: Quantum Mechanics, Feynman, Leighton, and Sands
The wave picture used in sections 3 and 4 (chapters 1, 16 and 17 for waves, amplitudes in position, and symmetry). [Online edition]
Quantum Mechanics: Amplitudes, Waves, and Spin, Alex Payne
The first set of notes in this series. These notes start where it stops: with a wave that has been confined and must therefore fit.

Optional reading and supportive texts

Modern Quantum Chemistry, Szabo and Ostlund
A standard text for Hartree–Fock, configuration interaction and coupled cluster (sections 7 and 10).

2 WHY THESE NOTES, AND WHAT THEY COVER

The first set of notes ended with stationary states: waves that keep their shape and only change phase, each with a definite energy. Any state is a sum of them. The next question is which stationary states to use for an electron bound to a nucleus, and what to do when the exact ones are not available, as they are not for any molecule. These notes follow that question to the labels that appear in the datasets used for machine-learned force fields, such as ωB97M-V/def2-TZVPD.

1. **Standing waves in a box** (section 3): a confined wave must fit, which gives whole numbers, energy levels and degeneracy.
2. **Spheres and atoms** (section 4): the same idea in spherical coordinates gives the spherical harmonics and the hydrogen orbitals.
3. **From atoms to molecules** (section 5): three approximations (fixed nuclei, one orbital per electron, atom-centred functions) and what each costs.
4. **Basis sets** (section 6): the fixed functions from which orbitals are built.
5. **Hartree–Fock** (section 7): the orbital approximation made precise.
6. **Density functional theory** (sections 8 and 9): the Kohn–Sham equations, and how to read a functional/basis-set label.
7. **Coupled cluster** (section 10): the reference method, and why it is the reference.

Units are atomic units in sections 4 to 10: lengths in bohr ($a_0 = 0.0529$ nm), energies in hartree (1 hartree $= 27.2$ eV), $\hbar = m_e = 1$. Each section is a condensed version of an interactive notebook in the `qm-theory` repository (the notebook names are given at the start of each section); numbers quoted below were computed there, and the last subsection of the technical notes lists what was and was not checked.

3 STANDING WAVES IN A BOX

Notebook: `standing_waves_in_boxes.py`.

3.1 *One dimension: whole numbers of half-wavelengths*

An electron in a box of length L has a wave function that must be zero at both walls, $\psi(0) = \psi(L) = 0$. The wave $\sin(kx)$ is zero at $x = 0$ automatically, and zero at $x = L$ only if kL is a whole multiple of π:

$$k_n = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots$$

That is, n half-wavelengths fit exactly between the walls. The free-particle energy $E = \hbar^2 k^2 / 2m_e$ then gives

$$E_n = \frac{\hbar^2 k_n^2}{2m_e} = n^2 \, \frac{\pi^2 \hbar^2}{2m_e L^2} = n^2 E_1.$$

The levels are labelled by a whole number because the wave has to fit; they are not evenly spaced ($E_n \propto n^2$); and the lowest energy is not zero, because squeezing the wave into a box forces it to curve and the curvature costs energy ($E_1 \propto 1/L^2$).

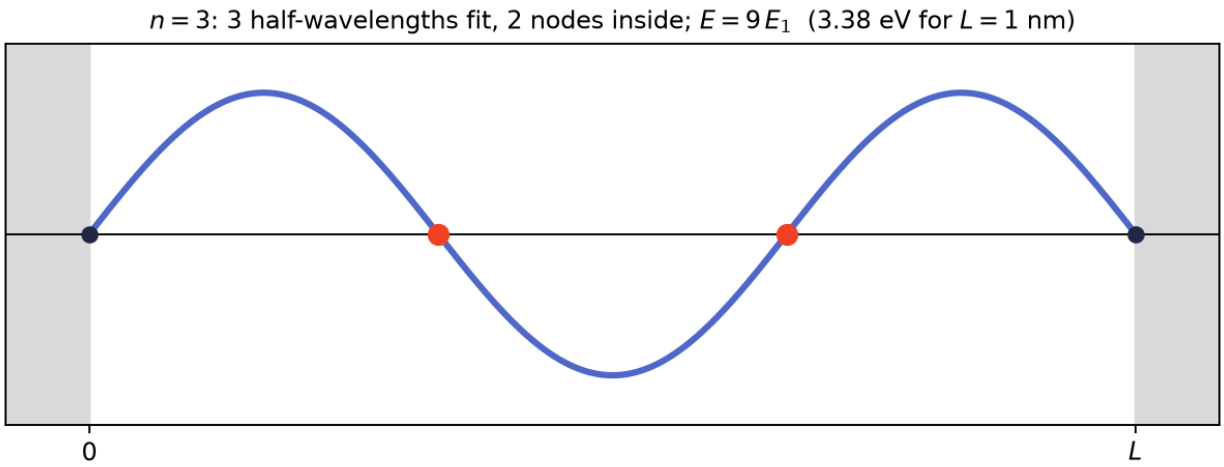


Figure 1: The wave with $n = 3$ half-wavelengths in a one-dimensional box.

3.2 *Two and three dimensions*

In a rectangular box with sides L_x and L_y the simplest waves that vanish on all walls are products, with a whole number per direction:

$$\psi_{n_x n_y} = \sin\!\left(\frac{n_x \pi x}{L_x}\right)\sin\!\left(\frac{n_y \pi y}{L_y}\right), \qquad E_{n_x n_y} = \frac{\pi^2 \hbar^2}{2m_e}\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2}\right).$$

The energies add because the kinetic energies of motion along x and y add. The wave changes sign across node lines: $n_x - 1$ cross the x direction and $n_y - 1$ the y direction. In three dimensions node lines become node planes and there are three whole numbers.

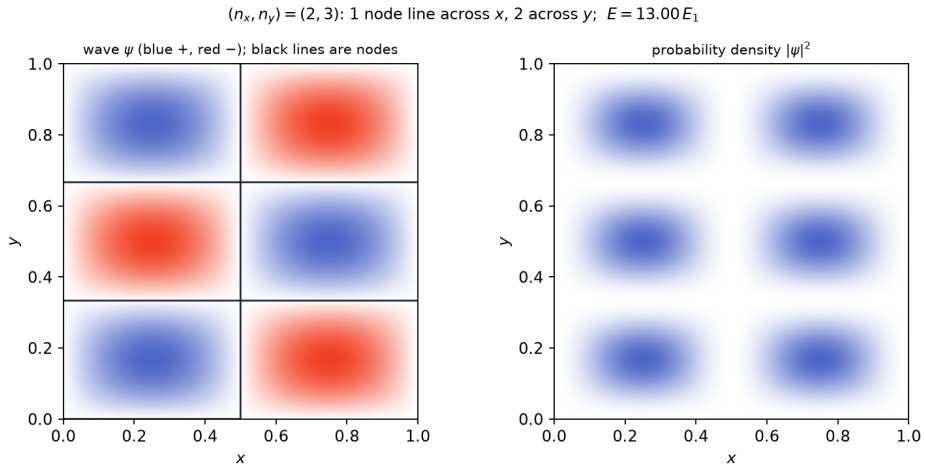


Figure 2: The wave with $(n_x, n_y) = (2, 3)$ in a square box; blue is positive, red negative.

3.3 *The allowed waves form a lattice, and energy is a distance*

With $\mathbf{k} = (k_x, k_y)$, $k_x = n_x \pi / L_x$, $k_y = n_y \pi / L_y$, the energy is $E = \frac{\hbar^2}{2m_e}|\mathbf{k}|^2$. The allowed waves are the points of a rectangular grid in the $\mathbf{k}$-plane, and *the energy of a state is the squared distance of its*

grid point from the corner. The walls decide which grid points exist; the distance decides the energy.

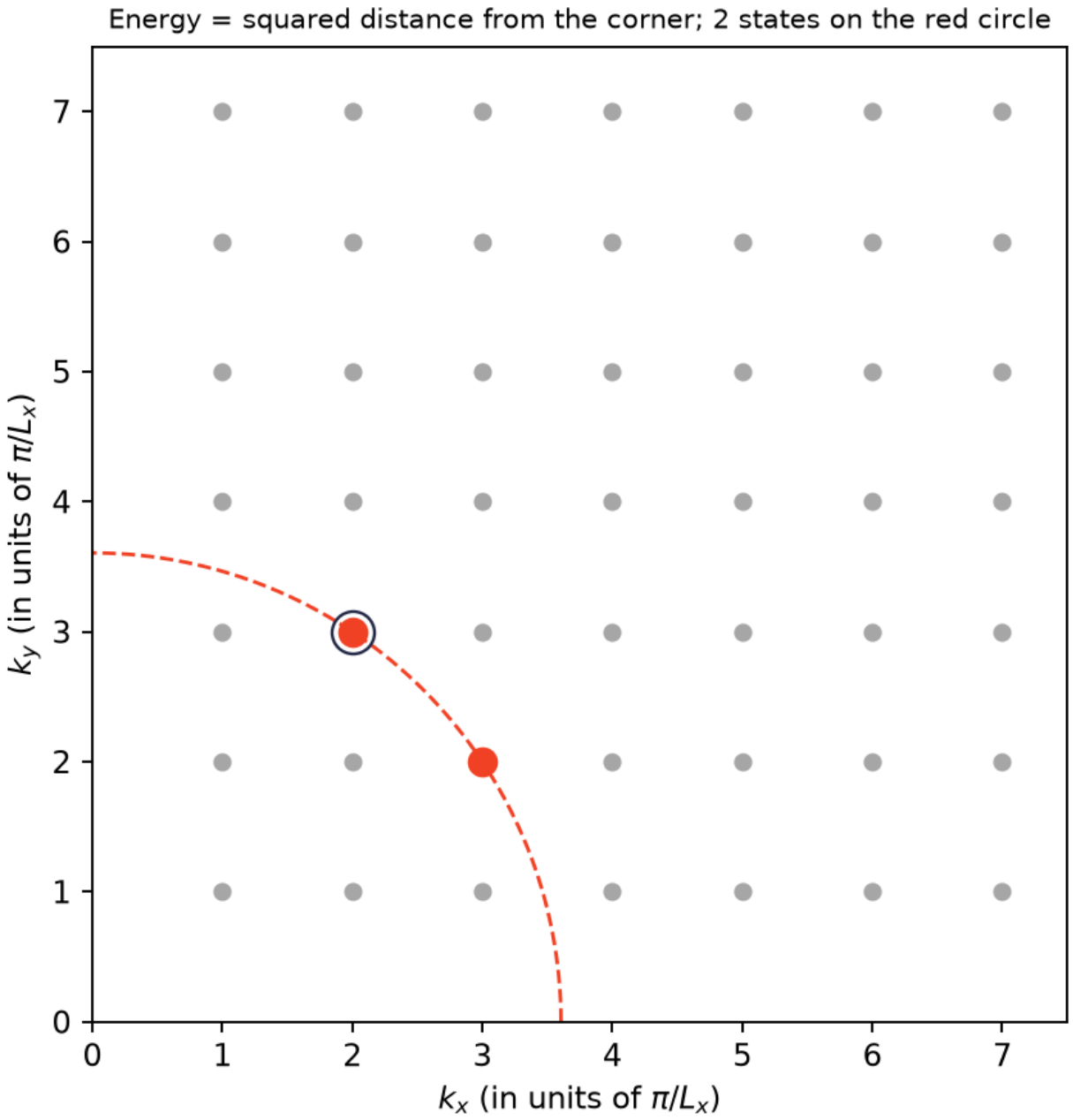


Figure 3: The allowed **k** values of a square box. States on one circle share an energy.

3.4 *Degeneracy is a sign of symmetry*

In a square box the states $(1,2)$ and $(2,1)$ are different waves with the same energy $5E_1$, because swapping x and y leaves the square unchanged. Some degeneracies have no such reason: $(1,7)$, $(5,5)$ and $(7,1)$ all have $50E_1$ because $1+49 = 25+25$ (an accidental degeneracy of whole numbers). Making one side longer removes the symmetry and splits the levels. In a cube the lowest levels have degeneracies $1, 3, 3, 3, 1, 6$, most of them from permuting the three axes.

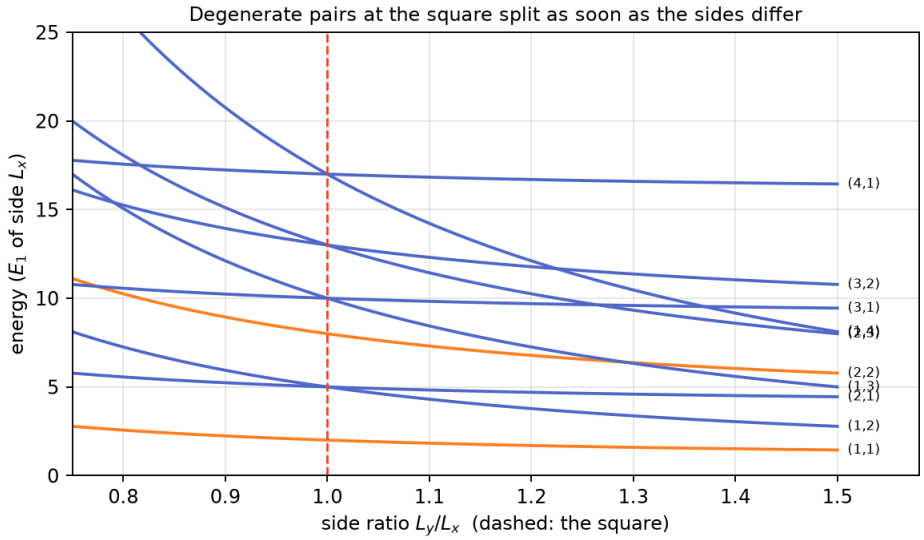


Figure 4: The lowest levels of a rectangular box as the ratio of the sides changes. At ratio 1 (the square) the symmetry gives degeneracies.

A sphere is the most symmetric container, and the degeneracy of the hydrogen levels (n^2 states in level n) follows from that.

3.5 *Counting states*

The number of states below E is the number of grid points inside a quarter circle (2D) or one eighth of a sphere (3D); for large E it is close to the area or volume. With E in units of E_1,

$$N_{1D} \approx \sqrt{E}, \qquad N_{2D} \approx \frac{\pi}{4}E, \qquad N_{3D} \approx \frac{\pi}{6}E^{3/2}.$$

More dimensions give many more states at the same energy, and the levels of a large box or a many-electron system merge into an almost continuous band. At atomic sizes the confinement energy is large: $E_1 \propto 1/L^2$ is tens of electronvolts for a box of about one bohr.

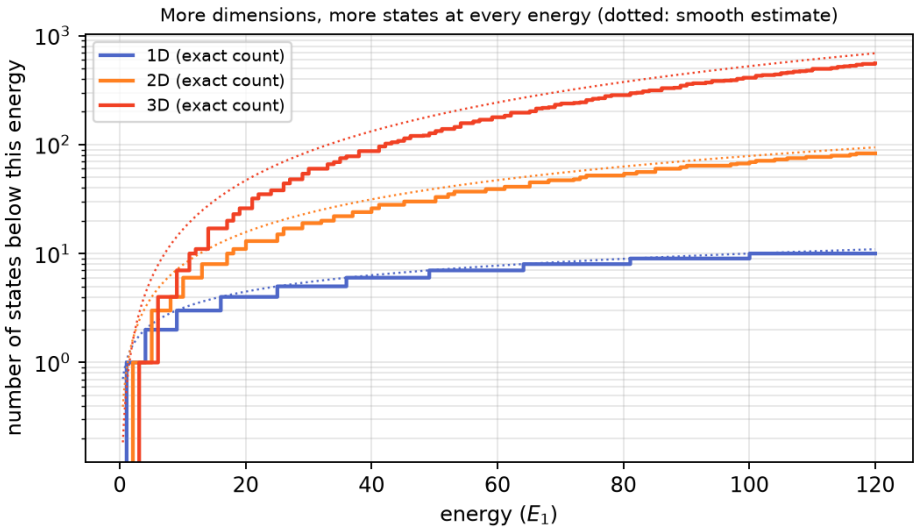


Figure 5: Number of states below an energy E in one, two and three dimensions.

FROM A BOX TO AN ATOM. An atom has no hard wall: the electron is held by the pull of the nucleus. Three things carry over: the wave must fit its surroundings (only some waves are allowed), each independent direction brings one whole number, and symmetry produces degeneracy. What changes is the choice of directions, which are no longer x, y, z but distance r, polar angle ϑ and azimuth φ.

4 SPHERES AND ATOMS: THE HYDROGEN ORBITALS

Notebook: `spherical_harmonics_and_hydrogen.py`.

4.1 *Spherical coordinates*

The natural coordinates for a potential that depends only on the distance from the nucleus are r, the polar angle ϑ (from the z axis) and the azimuth φ (around it). Volume elements change accordingly: $dV = dx\,dy\,dz$ becomes $r^2 \sin\vartheta\, dr\, d\vartheta\, d\varphi$, and the Laplacian becomes

$$\nabla^2 = \partial_r^2 + \frac{2}{r}\partial_r + \frac{\Lambda}{r^2},$$

where Λ contains only the angles. Because the potential depends on r alone, the variables separate: $\psi(r, \vartheta, \varphi) = R(r)\, Y(\vartheta, \varphi)$.

4.2 *Nodes and direction*

In a box, node points, lines and planes cut across the coordinate axes. In spherical coordinates the three families of nodes are *spheres* (constant r), *cones* (constant ϑ) and *planes through the axis* (constant φ). Writing $n_r, n_\vartheta, n_\varphi$ for the number of nodes in each family, the angular nodes number $\ell = n_\vartheta + n_\varphi$, and the total number of nodes is $N_{\text{nodes}} = n - 1 = n_r + n_\vartheta + n_\varphi$. The shape of an orbital is the pattern of these nodes.

4.3 *Standing waves on a sphere*

The angular part satisfies the eigenvalue equation

$$-\Lambda\, Y_{\ell m} = \ell(\ell+1)\, Y_{\ell m}.$$

Single-valuedness and finiteness of a wave on a sphere force m_ℓ and ℓ to be whole numbers, with $|m_\ell| \le \ell$. The $2\ell+1$ functions with one ℓ share the angular energy $\ell(\ell+1)$ (degeneracy from the spherical symmetry). The functions $Y_{\ell m}$ are the *spherical harmonics*; their real combinations give the familiar s, p, d shapes, with $|m_\ell|$ nodal planes through the axis and $\ell - |m_\ell|$ nodal cones.

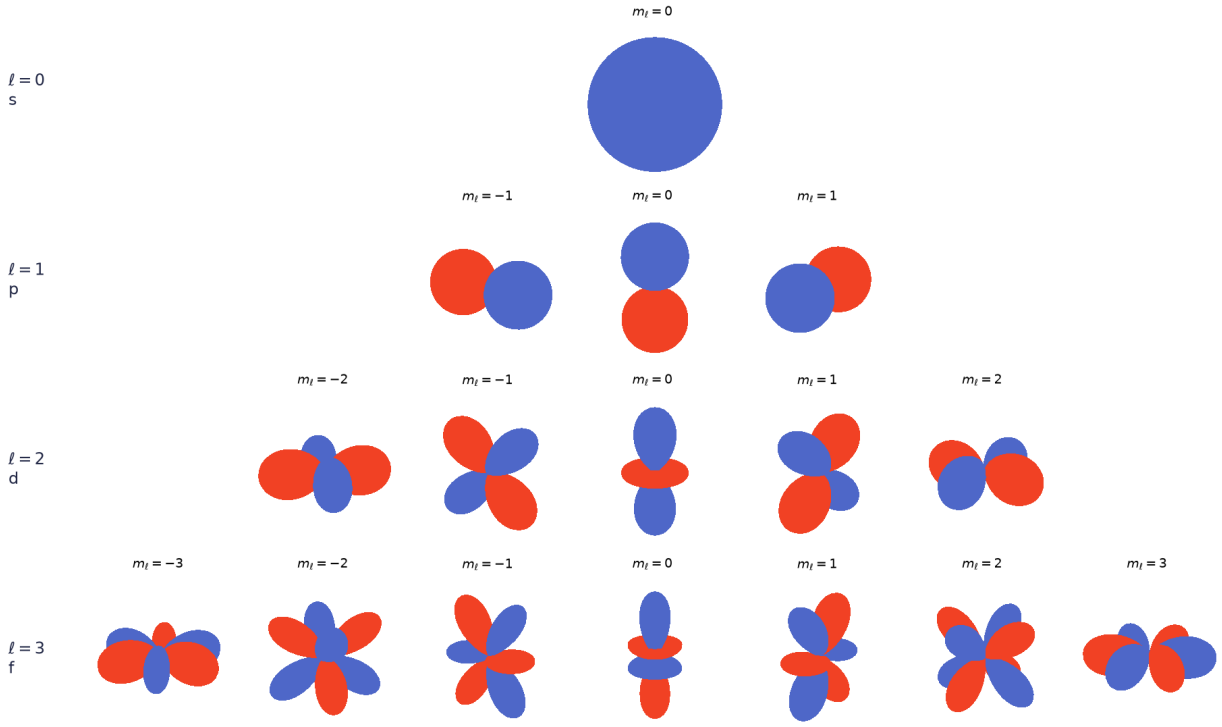


Figure 6: Real spherical harmonics for $\ell = 0$ to 3.

4.4 *The radial part*

The radial equation for $u = rR$ has the same form as the angular one, an eigenvalue problem, with $\ell(\ell+1)$ entering as a fixed number:

$$\Big[-\frac{1}{2}\frac{d^2}{dr^2} + \underbrace{\frac{\ell(\ell+1)}{2r^2} + V(r)}_{V_{\text{eff}}}\Big] u = E u.$$

The term $\ell(\ell+1)/2r^2$ is a barrier that keeps waves with $\ell > 0$ away from the nucleus. Each ℓ is therefore its own one-dimensional problem with its own ladder of energies, labelled by the number of radial nodes n_r. For hydrogen, $V = -1/r$, the energies are

$$E_n = -\frac{1}{2n^2}\ \text{hartree} = -\frac{13.6}{n^2}\ \text{eV}, \qquad n = n_r + \ell + 1.$$

The energy depends on n only, so level n has n^2 orbitals ($2n^2$ with spin). That the energy does not depend on ℓ is special to the potential $-1/r$; the oscillator and the drumhead differ, and so does every atom with more than one electron (section 5).

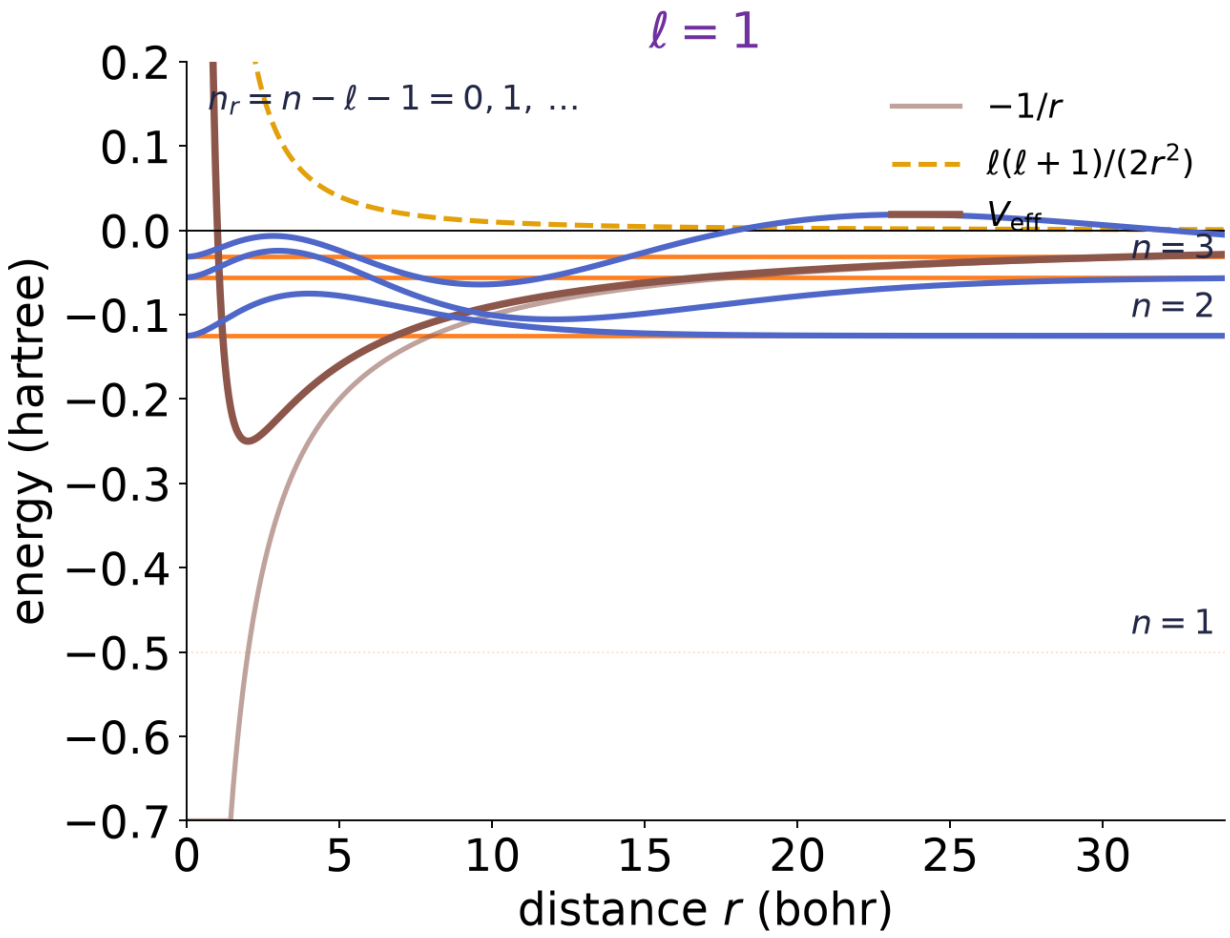


Figure 7: The effective potential for $\ell = 1$ (the barrier $\ell(\ell+1)/2r^2$ plus $-1/r$) and the lowest radial levels with their wave functions.

4.5 *Reading an orbital from its nodes*

The quantum numbers $(n_r, n_\vartheta, n_\varphi)$ say how many spheres, cones and planes the orbital has: a 3d orbital has $n = 3$, $\ell = 2$, so $N_{\text{nodes}} = 2$ and no radial node; a 4f has $N_{\text{nodes}} = 3$ and all three are angular. The energy counts N_{nodes}, the shape tells where they are.

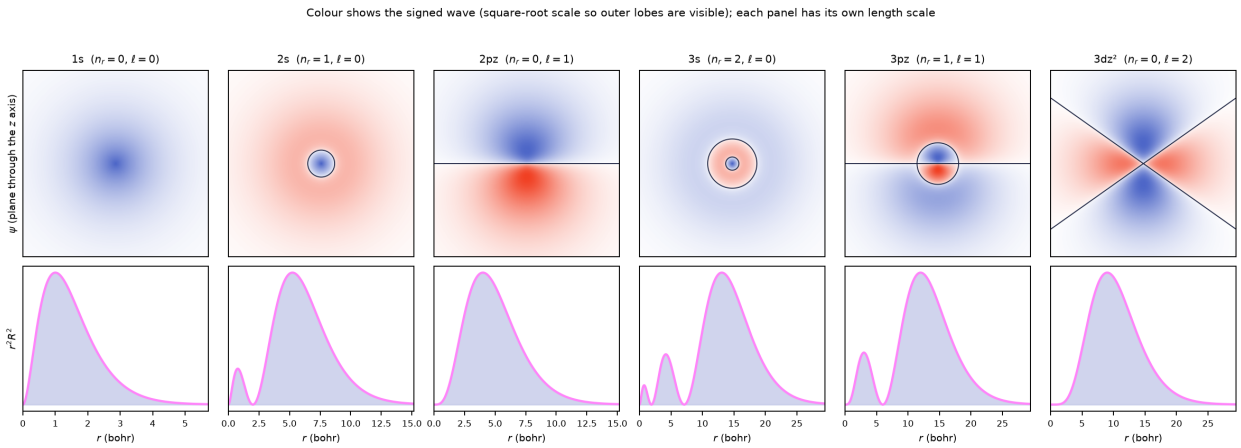


Figure 8: Sampled densities of hydrogen orbitals.

5 FROM ATOMS TO MOLECULES: THREE APPROXIMATIONS

Notebook: `from_atoms_to_molecules.py`.

Hydrogen is one of very few exactly solvable problems. A molecule has three complications: every particle adds three coordinates, the electrons push on each other through $1/|\mathbf{r}_i - \mathbf{r}_j|$, and there is no single centre of force. A table of the wave function on a grid with ten points per coordinate needs 10^N numbers for $N = 3(N_{\text{nuclei}} + N_{\text{electrons}})$ coordinates: 45 coordinates for methane, 378 for caffeine. Three approximations, made one after the other, make a calculation possible.

5.1 *Fixed nuclei: Born–Oppenheimer*

A proton is 1836 times heavier than an electron, so the electrons settle into their lowest arrangement for each position of the nuclei. The approximation (1) freezes the nuclei at positions $\mathbf{R}$ and solves for the electrons, giving an energy $E_{\text{el}}(\mathbf{R})$ that, with the nuclear repulsion, is a function of the nuclear positions only; and (2) uses that function as the potential for the much slower nuclear motion. The potential energy surface of every force field and geometry optimisation is this Born–Oppenheimer energy.

For H_2^+ (one electron, two protons a distance R apart) the minimum of $E(R)$ is at $R \approx 2.0$ bohr with a binding energy of about 2.8 eV. The nuclear problem is a one-dimensional standing-wave problem with reduced mass $\mu = M_p/2$,

$$\left[-\frac{1}{2\mu}\frac{d^2}{dR^2} + E(R) \right] \chi(R) = E_{\text{tot}}\,\chi(R),$$

whose levels have $0, 1, 2, \ldots$ nodes. The two lowest are 0.010 hartree (0.27 eV, about 2190 cm^{-1}) apart: about a tenth of the electron's binding energy, which is what "the electrons are fast" means in numbers. The spacing falls as $1/\sqrt{M}$, so lighter nuclei vibrate faster, and a heavier isotope changes the vibrational frequency but not the curve.

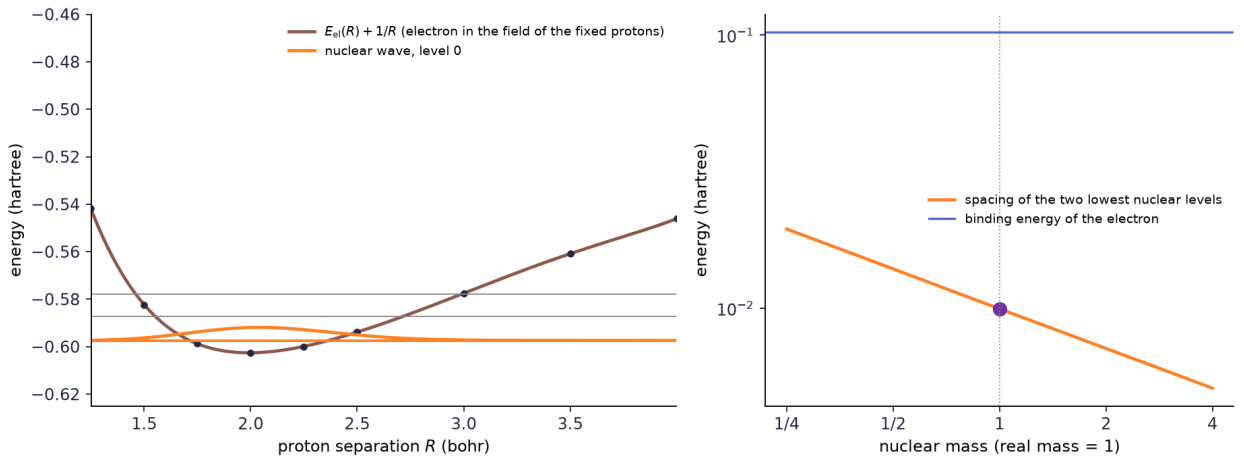


Figure 9: H_2^+: the electronic energy as a potential for the nuclei, the lowest nuclear levels (left), and the level spacing against nuclear mass (right).

5.2 *One orbital per electron*

The orbital approximation writes the many-electron wave as a product (made antisymmetric in section 7) of one orbital per electron, each electron moving in the field of the nucleus plus the *average* cloud of the others. In helium, take both electrons in the hydrogen-like function $e^{-\zeta r}$; the energy in hartree is

$$E(\zeta) = \underbrace{\zeta^2}_{\text{kinetic}} \underbrace{-2Z\zeta}_{\text{attraction}} + \underbrace{\tfrac{5}{8}\zeta}_{\text{electron repulsion}}, \qquad Z = 2.$$

Without the repulsion the minimum is at $\zeta = 2$ and $E = -4$. With it the minimum moves to $\zeta = Z - \frac{5}{16} = 1.6875$, $E = -2.8477$ hartree: the repulsion acts as a *screening*, and the orbital swells. The exact energy is -2.9037 and the best possible orbital product (the Hartree–Fock limit) is -2.8617. The orbital picture recovers 98 % of the energy, but the missing 0.042 hartree (1.1 eV) is larger than most chemical energy differences. It is the *correlation energy*: electrons dodge each other, which no product of orbitals can express.

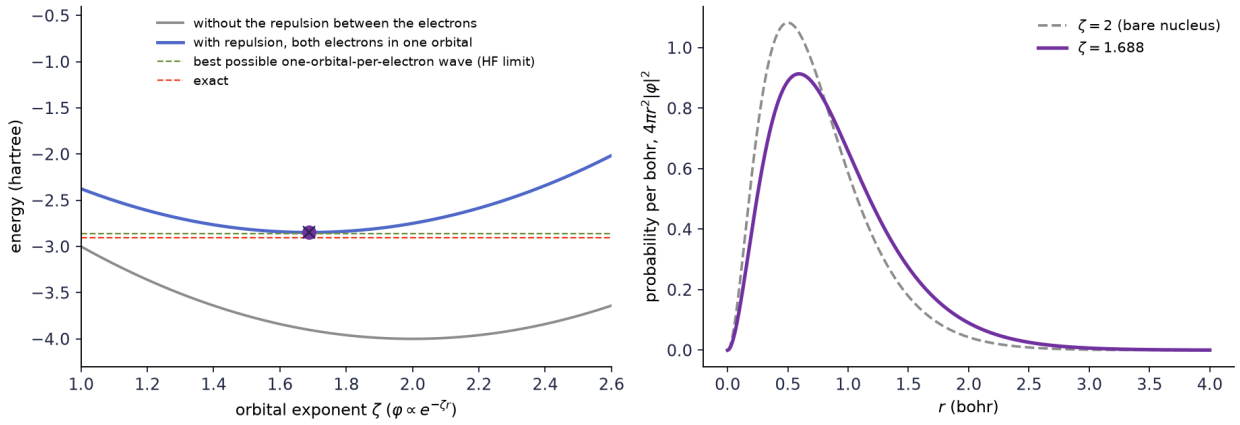


Figure 10: Helium with one shared 1s orbital: energy against the orbital exponent ζ (left) and the radial density of the orbital (right).

Screening also splits s from p at the same n. An inner electron screens the nucleus, so the effective charge seen by an outer electron is large close in and small far out. An s wave is large near the nucleus and feels the strong pull there; a p wave is kept away by the barrier $\ell(\ell+1)/2r^2$, feels the screened charge, and lies higher. In a toy model with a filled 1s shell, $Z_{\text{eff}}(r) = 1 + 2e^{-r/a}$, the order is $s < p < d$ at each n, and the splitting grows with a. This is the origin of the familiar filling order. (The toy has one parameter, set by hand rather than fitted to any atom, so only the ordering is a result.)

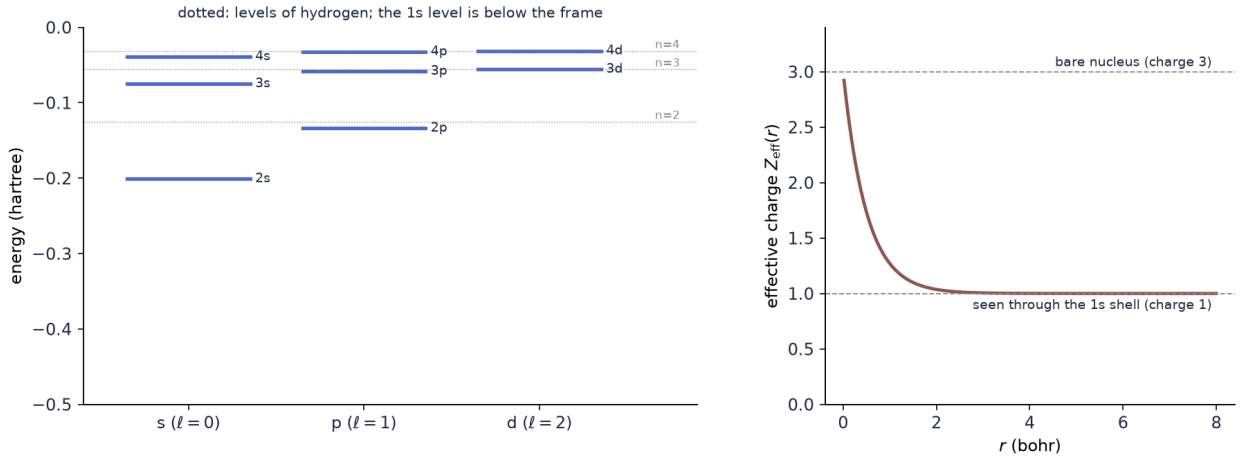


Figure 11: Levels of the toy atom with a filled 1s shell (left) and its effective charge (right).

5.3 *Hybrid orbitals*

An atom in a molecule may use any mixture of its own s and p orbitals; the energy of the whole occupied set does not change. A useful family points one lobe in direction $\hat{n}$:

$$h_{\hat{n}} = \frac{s + \sqrt{n}\, p_{\hat{n}}}{\sqrt{1+n}}, \qquad n = 1, 2, 3 \text{ for } sp, sp^2, sp^3.$$

Two such hybrids at angle θ overlap by $(1+n\cos\theta)/(1+n)$, and orbitals of one atom must be orthogonal, so

$$1+n\cos\theta=0 \implies \theta=\arccos\Big(-\tfrac{1}{n}\Big)=180°,\ 120°,\ 109.47°,$$

the angles of acetylene, ethene and methane. Four sp^3 hybrids $h_i=\frac{1}{2}\big(s+\sqrt{3}\,\hat{n}_i\cdot\mathbf{p}\big)$, with $\hat{n}_i$ the corners of a tetrahedron, are orthonormal and each has one quarter s character.

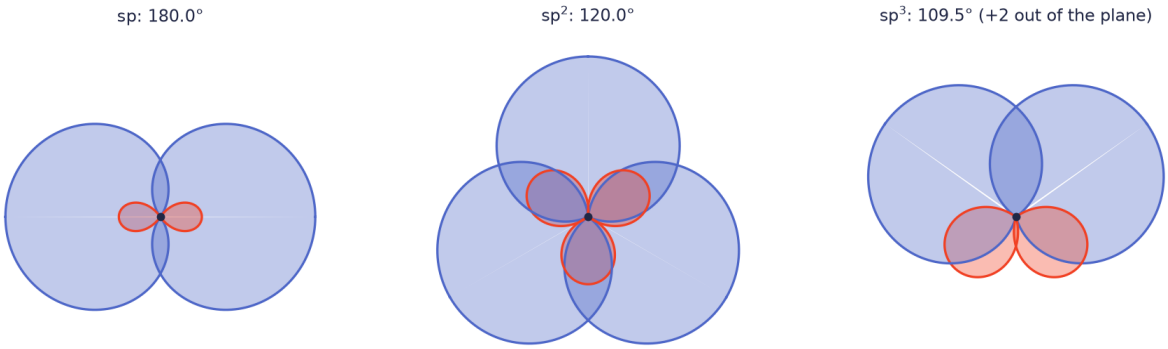


Figure 12: Hybrids in the plane that holds them; blue positive, red negative.

5.4 *Methane: one level, then three*

Take as basis the carbon 2s and three 2p functions and one 1s function on each hydrogen: eight functions, eight molecular orbitals. Symmetry reduces the problem to 2×2 blocks: the carbon 2s couples only to the sum of the four 1s functions (one level of symmetry a_1), and each carbon $2p_i$ couples only to the hydrogen combination pointing the same way (three identical levels, t_2). The four lower levels are bonding (one a_1, three t_2) and the four upper ones antibonding; methane's eight valence electrons fill the bonding ones. A real calculation (wB97M-V/def2-TZVPD) gives the same pattern: occupied orbital energies -10.319 (carbon 1s), -0.813 (a_1) and -0.493 hartree (three equal t_2).

The canonical orbitals are spread over the whole molecule. The four combinations $b_j=\frac{1}{2}\big(a_1+\sqrt{3}\,\hat{n}_j\cdot\mathbf{t}_2\big)$ span exactly the same occupied space and so the same electron density, but each sits on one C–H bond: hybrids are a rewriting of the occupied orbitals, not a physical effect. On carbon the real bond orbital is 20 % s, 76 % p and 4 % d, against 25 : 75 : 0 for ideal sp^3: real bonds lie near the symmetric ideal.

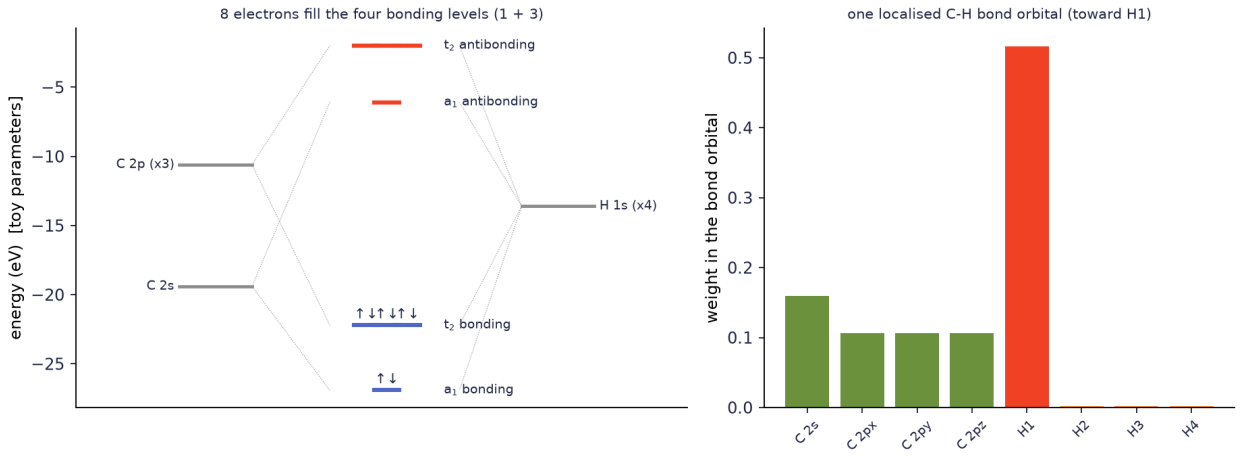


Figure 13: A symmetry toy for methane (invented parameters; only the pattern is a result): its levels (left) and the weights of one localised bond orbital (right).

6 BASIS SETS: FROM BASE STATES TO DEF2

Notebook: `basis_sets_from_base_states.py`.

6.1 *Base states, and what to do when they are unknown*

A stationary state ψ_E solves $H\psi_E = E\psi_E$, and a general wave is a sum $\psi = \sum_n c_n \psi_{E_n} e^{-iE_n t/\hbar}$ of these base states. For hydrogen the base states are the orbitals of section 4. For a molecule they are not known in closed form, so one builds the wave from other fixed functions χ_ν that one is free to choose, $\psi = \sum_\nu c_\nu \chi_\nu$, and lets the coefficients adjust. Putting the sum into $H\psi = E\psi$, multiplying by χ_μ and integrating gives a matrix equation,

$$\sum_\nu H_{\mu\nu} c_\nu = E \sum_\nu S_{\mu\nu} c_\nu, \qquad H_{\mu\nu} = \int \chi_\mu H \chi_\nu, \quad S_{\mu\nu} = \int \chi_\mu \chi_\nu,$$

which is the same as $\mathbf{Hc} = E\,\mathbf{Sc}$ (a generalized eigenproblem; $\mathbf{S} = \mathbf{1}$ for orthonormal functions). The variational principle says that the lowest energy from a truncated sum is never below the exact one, and never rises when a function is added.

6.2 *Many nuclei: atom-centred functions (LCAO)*

For molecules the functions are centred on the atoms, $\psi = \sum_\mu c_\mu \chi_\mu$ (LCAO, linear combination of atomic orbitals), because an electron near a nucleus mostly feels that nucleus. For H_2^+, with one hydrogen 1s function $e^{-r}/\sqrt{\pi}$ on each proton, symmetry gives the bonding combination $c_A = c_B$ and the antibonding combination $c_A = -c_B$, with energies $E_\pm = (H_{AA} \pm H_{AB})/(1 \pm S)$ plus the proton repulsion $1/R$. Only the bonding curve dips below the energy $-\frac{1}{2}$ of a separated hydrogen atom and a proton. Two functions put the minimum at $R \approx 2.49$ bohr with a binding energy of about 1.8 eV; a very large basis gives $R \approx 2.0$ bohr and about 2.8 eV, so two functions give roughly 63 % of the bond energy and lie above the large-basis curve at every R, as the variational principle requires. Making the basis richer on each atom is what the rest of this section is about.

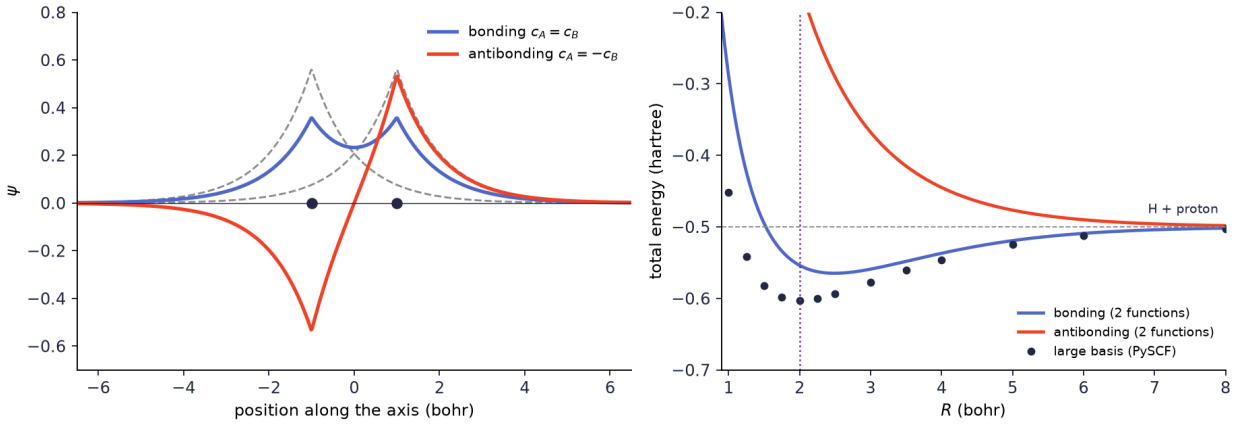


Figure 14: H_2^+ with one 1s function per proton: the bonding and antibonding combinations (left) and their energies, against a large-basis calculation (right).

6.3 *Slater and Gaussian functions*

The exact 1s function has a corner at the nucleus (the cusp, $\psi'(0) = -Z\psi(0)$) and a slow exponential tail, so a *Slater-type* function $e^{-\zeta r}$ is a good copy. A *Gaussian* $e^{-\alpha r^2}$ has neither: zero derivative at the centre and a tail that falls too fast. Gaussians are used anyway because the product of two Gaussians on different centres is a single Gaussian at a point between them,

$$e^{-\alpha(x-A)^2} e^{-\beta(x-B)^2} = K\, e^{-(\alpha+\beta)(x-P)^2}, \quad P = \frac{\alpha A + \beta B}{\alpha + \beta}, \quad K = e^{-\frac{\alpha\beta}{\alpha+\beta}(A-B)^2},$$

so the millions of integrals in **H** and **S** are known in closed form and fast. The shape is repaired by *contraction*: a fixed sum of n Gaussians used as one function (STO-nG fits a Slater function by n Gaussians).

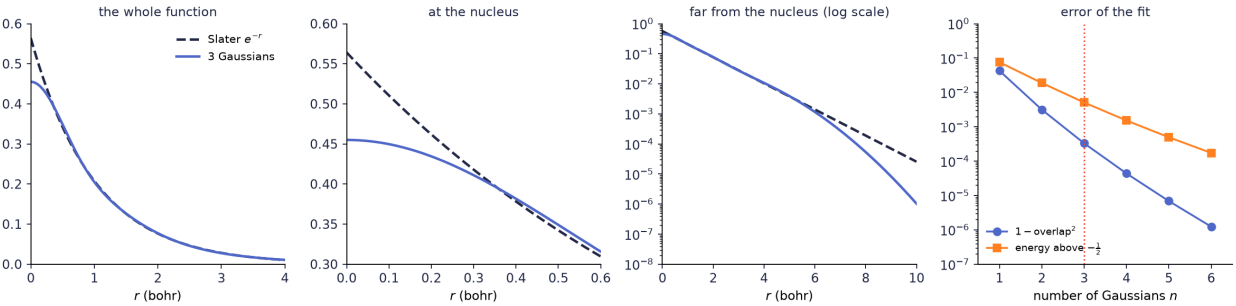


Figure 15: A Slater 1s function and its fit by three Gaussians.

6.4 *Shells, zeta, polarization and diffuse functions*

A basis function is a radial part times a real spherical harmonic. A *shell* of angular momentum ℓ is the set of $2\ell + 1$ functions with one radial part: 1 for s, 3 for p, 5 for d, 7 for f. Several kinds of shells are combined:

- **Zeta** (ζ): the valence shell is described by 2, 3 or 4 functions of different width (double, triple, quadruple zeta), so the orbital can swell or shrink in a bond.
- **Polarization** (P): shells of higher angular momentum than the atom's occupied ones (d on carbon, p on hydrogen). Adding some p_z to an s function shifts the cloud off-centre, as a bond requires.
- **Diffuse** (D): very wide, low-exponent functions for the far region, needed for anions, hydrogen bonds and dispersion.

In the label `def2-TZVPD`: `def2` is one consistent family of Gaussian basis sets (designed by Weigend and Ahlrichs), `TZ` is triple zeta, `V` says the extra functions are for the valence shells, `P` polarization, `D` diffuse. Water has 58 basis functions in `def2-TZVPD` and 74 in `def2-TZVPPD`, which differs only on hydrogen (one more p shell and a d shell).

7 HARTREE–FOCK: THE BRIDGE TO DFT

Notebook: `reading_a_dft_label.py`, *section 2.*

With N electrons the repulsion $1/r_{12}$ depends on two positions at once. The first practical approximation gives each electron *its own orbital* ψ_i and lets it feel only the average of the others. Two ideas turn this into a method.

ANTISYMMETRY. Electrons are fermions: swapping two of them must flip the sign of Ψ (the previous notes show where this comes from). A product of orbitals does not do this, but a determinant does:

$$\Psi(1,\ldots,N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(1) & \cdots & \psi_N(1) \\ \vdots & & \vdots \\ \psi_1(N) & \cdots & \psi_N(N) \end{vmatrix}.$$

Swapping two electrons swaps two rows, which flips the sign. Two electrons in the same orbital make two columns equal, so $\Psi = 0$: the Pauli principle comes out for free.

THE VARIATIONAL PRINCIPLE. Any trial wave function has an energy at or above the true ground-state energy, so the best orbitals are those that make the energy of the determinant as low as possible. This gives the Hartree–Fock (HF) equations,

$$\hat{f}\psi_i = \varepsilon_i \psi_i, \qquad \hat{f} = -\tfrac{1}{2}\nabla^2 + v_{\text{nuc}} + \hat{J} - \hat{K},$$

where $\hat{J}$ is the classical repulsion from the average density and $\hat{K}$ is *exchange*, a correction with no classical counterpart that comes entirely from antisymmetry. It removes the unphysical self-repulsion that $\hat{J}$ alone contains and keeps electrons of the same spin apart. Written in a basis set, the equations become the matrix equation $\mathbf{FC} = \mathbf{SC}\varepsilon$ of the next section.

WHAT HARTREE–FOCK MISSES. Each electron sees only the *average* position of the others; real electrons dodge each other, and the energy this saves is the correlation energy. It is a small fraction of the total, but of the size of the energy differences chemistry cares about. Dispersion, the attraction between two uncharged molecules, is a pure correlation effect, so HF cannot produce it: HF finds two methane molecules repulsive. Two ways forward exist: wave-function methods (MP2, coupled cluster; section 10), accurate but with steeply rising cost, and density functional theory (next), with a cost close to HF's but an approximate exchange–correlation term.

8 DENSITY FUNCTIONAL THEORY AND THE KOHN–SHAM MATRIX

Notebooks: `reading_a_dft_label.py`, *section 3, and* `kohn_sham_matrix.py`.

For N electrons the wave function depends on 3N coordinates; DFT replaces it by the density $\rho(\mathbf{r})$, which depends on three. The Hohenberg–Kohn theorem says the ground-state energy is determined by ρ alone, so there is an energy functional $E[\rho]$. Nobody knows it exactly, but Kohn and Sham split it so that only a small piece is unknown: take non-interacting electrons that happen to have the same density as the real ones, with orbitals φ_i,

$$\rho(\mathbf{r}) = \sum_{i}^{\text{occ}} |\varphi_i(\mathbf{r})|^2, \qquad E[\rho] = T_s[\rho] + E_{\text{nuc}}[\rho] + E_H[\rho] + E_{xc}[\rho].$$

T_s is the kinetic energy of the non-interacting electrons, E_{nuc} the attraction to the nuclei, E_H the classical repulsion of the density with itself, and E_{xc} gathers everything left over (exchange and correlation). *The "functional" in a DFT label is a specific approximation to* E_{xc}*;* everything else in the sum is exact.

8.1 *The matrix equation*

The orbitals satisfy $\hat{f}\psi_i = \varepsilon_i \psi_i$ with $\hat{f} = -\frac{1}{2}\nabla^2 + v_{\text{eff}}[\rho]$. Writing each orbital as a sum of N fixed basis functions, $\psi_i = \sum_\mu C_{\mu i}\phi_\mu$, inserting it, multiplying by ϕ_ν and integrating (as in section 6) gives

$$\boxed{\mathbf{FC} = \mathbf{SC}\varepsilon}, \qquad F_{\mu\nu} = \int \phi_\mu \hat{f} \phi_\nu \, d\mathbf{r}, \quad S_{\mu\nu} = \int \phi_\mu \phi_\nu \, d\mathbf{r}.$$

Column i of $\mathbf{C}$ holds the weights of orbital i, and ε is the diagonal matrix of orbital energies. N basis functions give N orbitals: for water in `def2-TZVPD`, 58 orbitals, of which the 5 lowest hold the 10 electrons and the other 53 are empty (virtual). Basis functions (indices μ, ν) are chosen before the calculation; orbitals (indices i, j) are its result.

8.2 *What is in* $\mathbf{F}$

$$\mathbf{F} = \mathbf{T} + \mathbf{V} + \mathbf{J}[\mathbf{D}] + \mathbf{V}^{xc}[\mathbf{D}] + a\,\mathbf{K}[\mathbf{D}],$$

where $\mathbf{D} = 2\mathbf{C}_{\text{occ}}\mathbf{C}_{\text{occ}}^{\mathsf{T}}$ is the density matrix. $\mathbf{T}$ (kinetic) and $\mathbf{V}$ (nuclear attraction) depend only on geometry and basis; $\mathbf{J}$ is the Coulomb repulsion of an electron with the whole density; $\mathbf{V}^{xc}$ is the exchange–correlation term, evaluated on a numerical grid; $a\mathbf{K}$ is the fraction a of exact (Hartree–Fock) exchange that a hybrid functional mixes in ($a = 0$ pure functional; $a = 1$ with no $\mathbf{V}^{xc}$ is Hartree–Fock itself); and the "-V" in ωB97M-V adds one more, non-local term. For the first element F_{11} of water at ωB97M-V/`def2-TZVPD` the pieces are (hartree) $T = +76.978$, $V = -99.580$, $J = +19.827$, $V^{xc} = -3.611$, VV10 $+0.005$ and exact exchange -1.024, adding up to -7.404. The big terms T, V and J nearly cancel; exchange–correlation is a small correction, which is why its approximation matters so much for energy *differences*.

Everything except $\mathbf{T}$ and $\mathbf{V}$ depends on $\mathbf{D}$, which comes from solving with $\mathbf{F}$. That circularity is why the self-consistent-field (SCF) loop is repeated:

1. guess a density;
2. build $\mathbf{F}$ from it;
3. solve $\mathbf{FC} = \mathbf{SC}\varepsilon$;

4. form the new density and repeat until the energy stops changing.

9 READING A DFT LABEL

Notebook: `reading_a_dft_label.py`.

The two labels of the machine-learning datasets used in the lab are

	OMol25	SPICE II
Functional	ωB97M-V	ωB97M
Dispersion	part of the functional (VV10)	added afterwards (D3 with BJ damping)
Basis set	`def2-TZVPD`	`def2-TZVPPD`
Written as	ωB97M-V/`def2-TZVPD`	ωB97M-D3(BJ)/`def2-TZVPPD`

Three choices define the level of theory: the **functional** (how the electron interaction is approximated), the **dispersion treatment** (how the weak attraction between neighbouring molecules is described) and the **basis set** (what shapes the orbitals are built from). Underneath are settings that are usually *not* in the label but still change the number: the integration grid, the convergence thresholds and the program.

9.1 *The basis set and why the D matters*

When two molecules are close, each can "borrow" the other's basis functions to describe its own electrons better, and the energy drops for a reason unrelated to chemistry: the basis-set superposition error (BSSE). The *counterpoise* correction recomputes each monomer in the full dimer basis, using ghost atoms (basis functions without nuclei or electrons). A molecule gains energy by borrowing only if its own basis is missing something: the faint tail of its cloud that reaches toward the neighbour. A diffuse function is still centred on its own atom, but its small exponent lets it describe that tail, so there is nothing left to borrow. Most of the drop in BSSE from `def2-TZVP` to `def2-TZVPD` is exactly this. Anions need diffuse functions for the same reason: the extra electron is loosely bound.

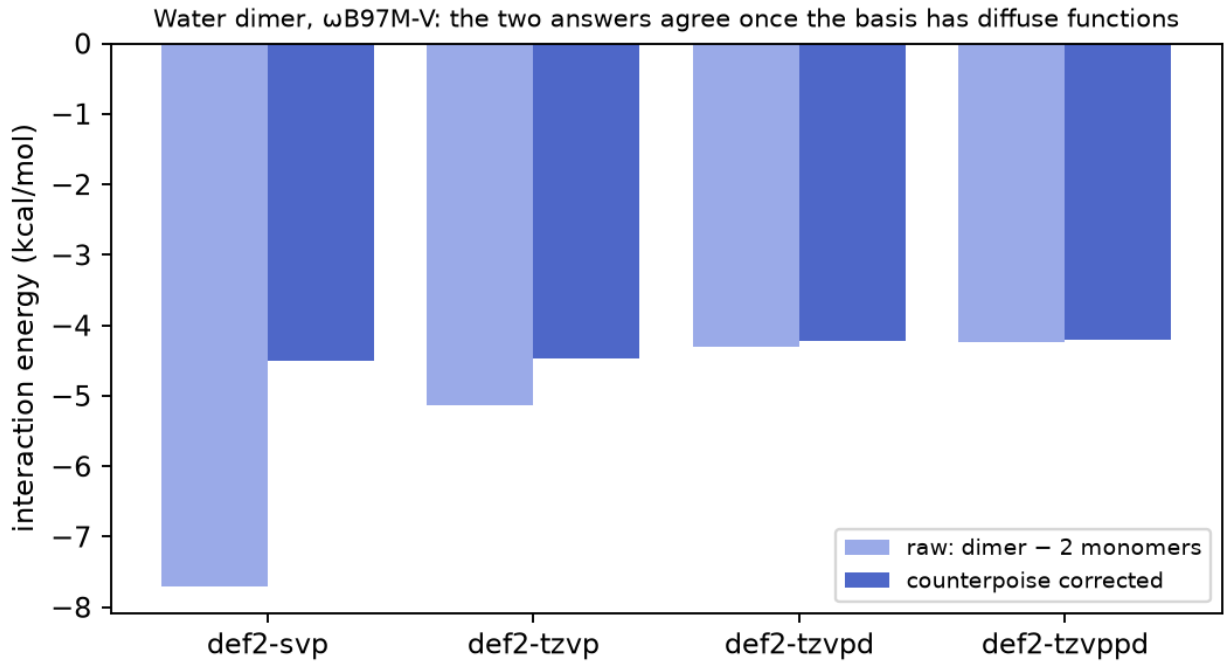


Figure 16: Water dimer at ωB97M-V: interaction energy with and without the counterpoise correction, for several basis sets.

9.2 *The functional: Jacob's ladder*

Approximations to E_{xc} can be sorted by what they are allowed to look at, as the rungs of a ladder:

Rung	Looks at	Examples
1. LDA	the density ρ at each point	SVWN
2. GGA	ρ and its gradient $\lvert\nabla\rho\rvert$	PBE, B97
3. meta-GGA	also the kinetic energy density τ	r2SCAN, ωB97M (the M)
4. hybrid	also some exact (HF) exchange from the orbitals	B3LYP, ωB97X, ωB97M
5. double hybrid	also virtual orbitals	(not used for these datasets)

B97 names the form of the GGA part: Becke's 1997 functional writes exchange and correlation as power series in the reduced gradient with fitted coefficients, and the members of the family (B97, ωB97, ωB97X, ωB97X-V, ωB97M-V) share a shape and differ in how the coefficients were chosen. **M** means meta: the functional also uses $\tau(\mathbf{r}) = \frac{1}{2}\sum_i |\nabla\varphi_i|^2$. Compared with its value τ_W for a single orbital, τ tells the functional what kind of region it is in: $\tau \approx \tau_W$ where one orbital dominates (far outside the molecule, in the oxygen 1s core), $\tau > \tau_W$ where several overlap (the bonding region). A GGA cannot make this distinction; a meta-GGA can.

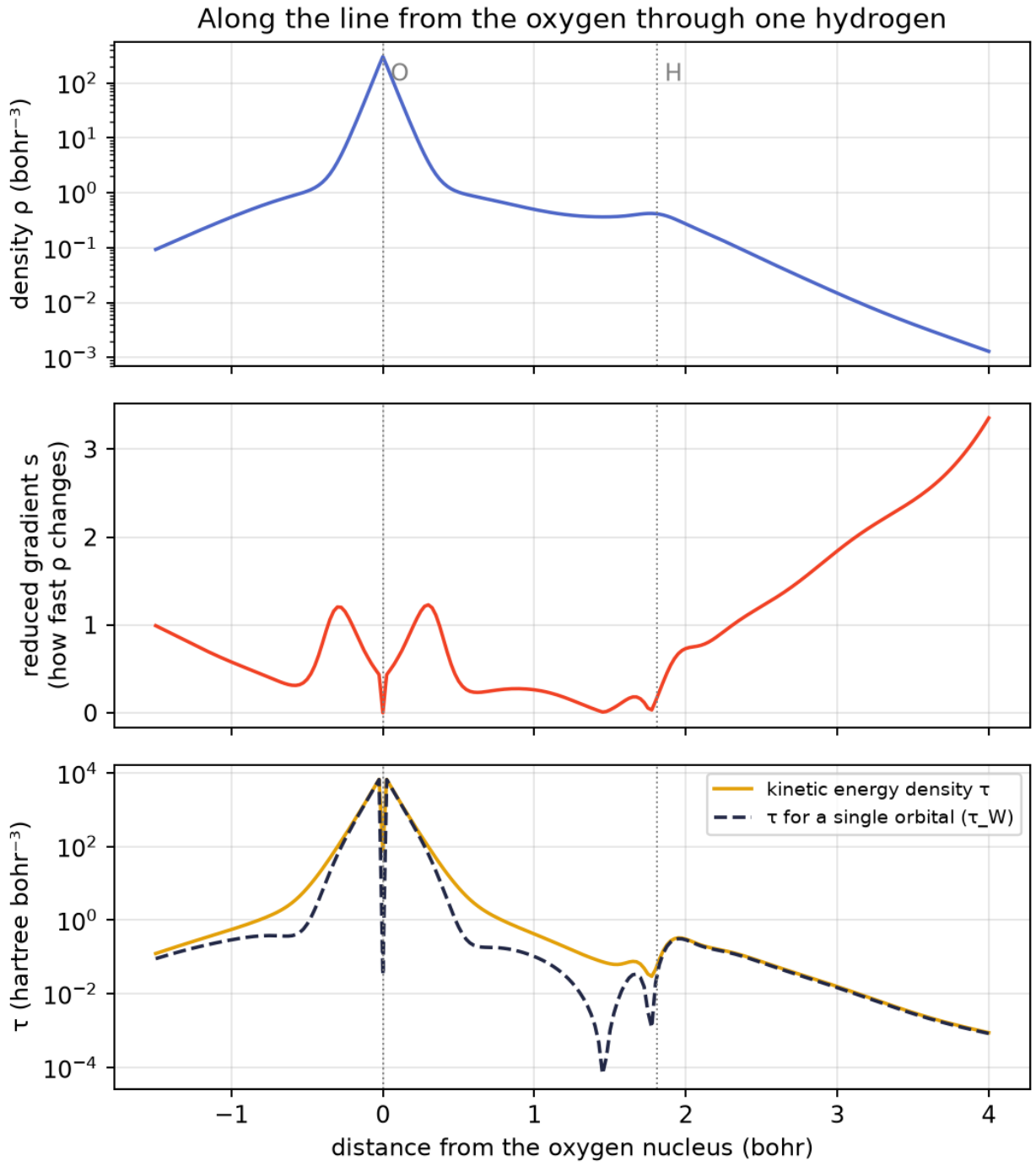


Figure 17: The ingredients a functional looks at, on a line from the oxygen nucleus through one hydrogen of water.

9.3 *Exact exchange and range separation: ω*

Exchange between electrons of the same spin is known exactly from Hartree–Fock theory; in DFT it is approximated, with a known flaw: an electron partly repels itself (self-interaction error). Mixing in some exact exchange gives a hybrid (B3LYP uses 20 % everywhere). A *range-separated* hybrid is

smarter about where: it splits the repulsion in two,

$$\frac{1}{r} = \underbrace{\frac{\operatorname{erfc}(\omega r)}{r}}_{\text{short range}} + \underbrace{\frac{\operatorname{erf}(\omega r)}{r}}_{\text{long range}},$$

where ω is the splitting parameter (unrelated to angular frequency) and r the distance between two electrons. Up close the density functional does the job; far away the exact $1/r$ tail matters, so the long-range part uses 100 % exact exchange. For ωB97M-V, $\omega = 0.3$ bohr^{-1} with 100 % long-range and 15 % short-range exact exchange. A test of why this matters: for the exact functional the energy of the highest occupied orbital equals minus the ionisation energy. Local functionals and B3LYP miss by 3 to 5 eV, because their wrong tail lets the outermost electron drift away too easily; range-separated hybrids land within about 1 eV (the experimental value for water is 12.6 eV).

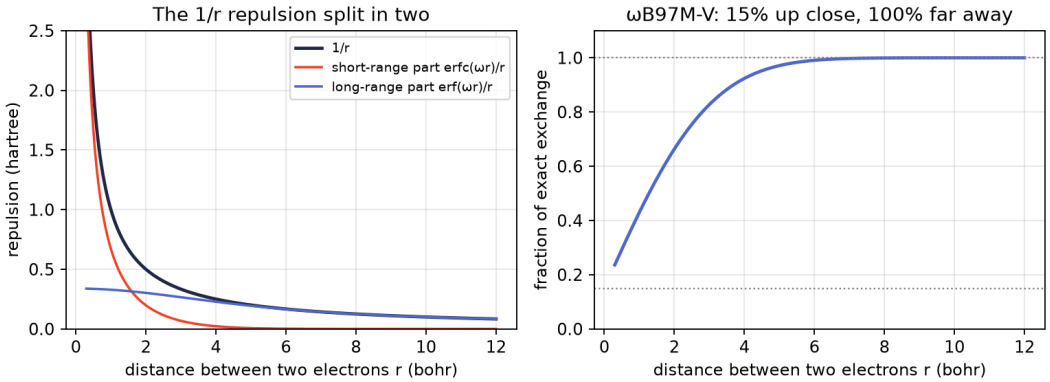


Figure 18: The short- and long-range parts of $1/r$ and the exact-exchange fractions of ωB97M-V.

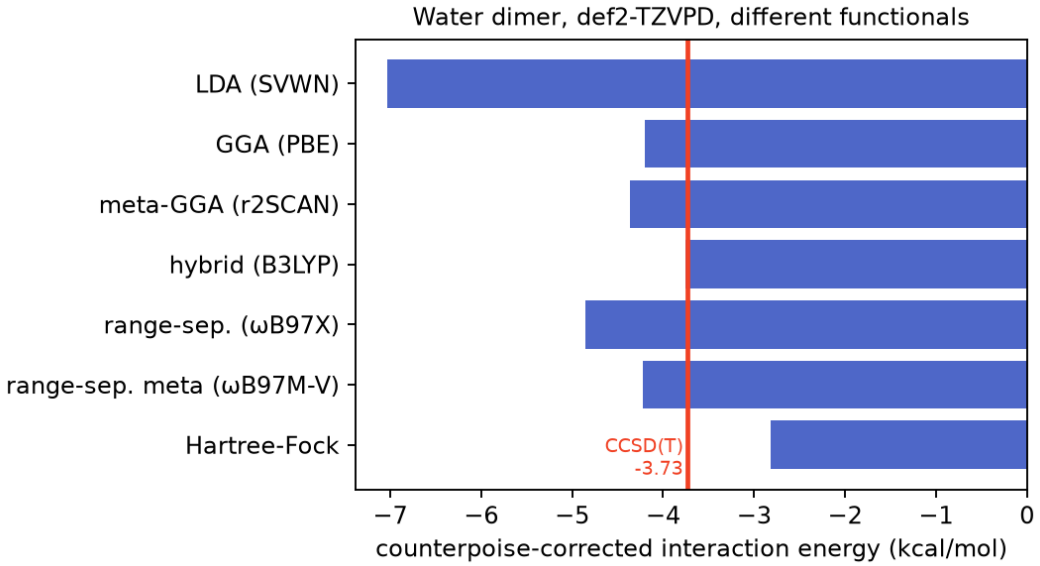


Figure 19: Water dimer interaction energy (counterpoise corrected, `def2-TZVPD`) for several methods, against coupled cluster in the same basis.

The reference is itself limited by the basis, so the bars are a comparison, not a ranking of truth.

Local functionals over-bind the hydrogen bond, Hartree–Fock under-binds it, B3LYP lands closest (partly through cancelling errors, since it has no dispersion term), and ωB97M-V binds about 0.5 kcal/mol more than the reference.

9.4 *Dispersion: -V against D3(BJ)*

London dispersion is an attraction between instantaneous fluctuations of the electron clouds of two molecules, falling as $-C_6/r^6$. It is *non-local*: it connects density in one place with density in another, so local functionals and hybrid exchange cannot produce it. Two remedies are the difference between the two labels:

- **-V (VV10)** adds a non-local correlation term computed from the density on the grid; the functional's parameters were fitted together with it, so it is part of ωB97M-V.
- **-D3(BJ)** adds an empirical pairwise term $-\sum C_6/r^6 + \cdots$ after the electronic calculation, with tabulated C_6 coefficients; the Becke–Johnson (BJ) damping function switches it off at short distance, where the functional already describes the electrons and the bare $-C_6/r^6$ would diverge.

The test is two methane molecules 3.8 Å apart, which have no hydrogen bond or charge, so whatever binding they have is dispersion. Hartree–Fock and B3LYP find them repulsive; ωB97M-V without VV10 leaves almost nothing; with VV10 the dimer binds by 0.42 kcal/mol, close to the coupled-cluster value 0.39.

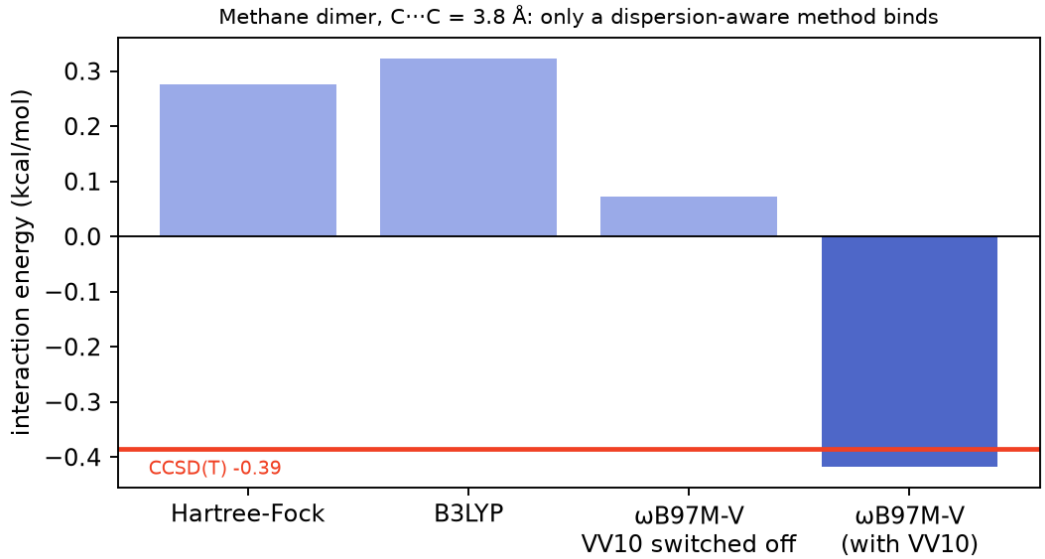


Figure 20: Methane dimer at 3.8 Å: interaction energy for several methods against coupled cluster.

9.5 *Settings that are not in the label*

DFT integrals of E_{xc} are done numerically on a grid of points around each atom, specified as (radial, angular) points; the water calculations in these notes use $(99, 590)$. Meta-GGAs and non-local terms are more sensitive to the grid than plain GGAs. The grid, the convergence criteria and the program (PySCF, ORCA, Psi4, Q-Chem) are not part of the label, so two groups with the "same" level of theory may differ by small amounts.

10 COUPLED CLUSTER: WHY CCSD(T) IS THE REFERENCE

Notebook: `coupled_cluster.py`.

10.1 *The exact answer, and why it is too expensive*

Hartree–Fock gives the best single determinant Φ_0 of N occupied spin orbitals and, in a basis of M spatial orbitals, also $2M - N$ virtual ones. Every way of moving electrons from occupied to virtual orbitals gives another determinant: Φ_i^a (one electron moved, a single), Φ_{ij}^{ab} (two, a double), and so on. Together they span every antisymmetric wave function in the basis, and the exact answer in that basis is a combination of all of them,

$$\Psi_{\mathrm{FCI}} = c_0\Phi_0 + \sum c_i^a \Phi_i^a + \sum c_{ij}^{ab}\Phi_{ij}^{ab} + \sum c_{ijk}^{abc}\Phi_{ijk}^{abc} + \cdots ,$$

found by diagonalising the Hamiltonian between all determinants (full configuration interaction, FCI). The correlation energy in the basis is $E_{\mathrm{FCI}} - E_{\mathrm{HF}}$. The number of determinants is $\binom{M}{N_\alpha}\binom{M}{N_\beta}$, which grows exponentially: for water ($N_\alpha = N_\beta = 5$) it is 1.7 million in the 6-31G basis, about 10^9 in cc-pVDZ and about 10^{13} in cc-pVTZ. For water in the smallest basis, where FCI is feasible, the reference determinant carries most of the wave function and the doubles almost all of the remainder; singles, triples and quadruples together are a small fraction. So most of the exact answer is the reference plus pairs of electrons moved together.

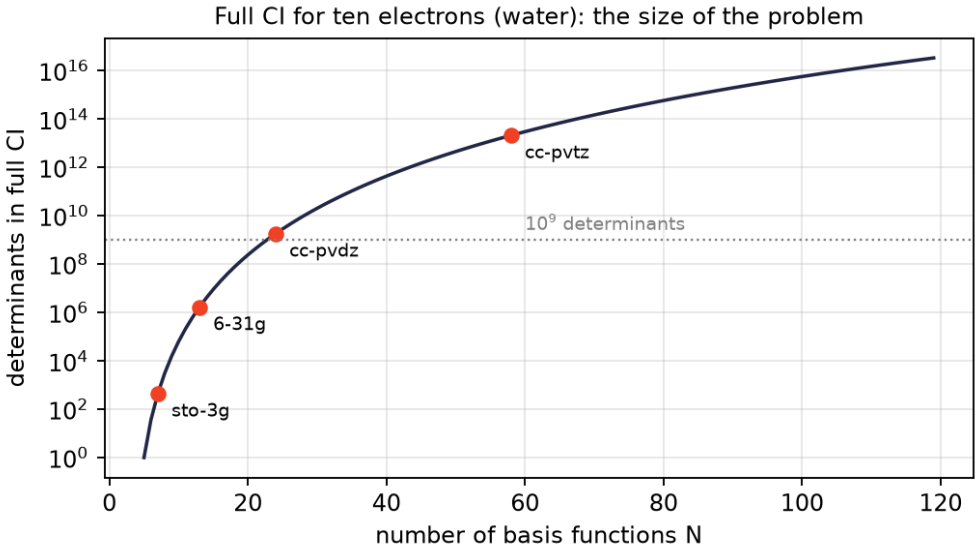


Figure 21: Number of determinants in FCI for water, by basis set.

10.2 *Truncating the exact answer breaks size consistency*

The obvious cut keeps determinants up to doubles: CISD, with coefficients from the lowest eigenvector of the Hamiltonian in that smaller space. It is variational and cheap, but it is not *size consistent*: the energy of two molecules infinitely far apart should equal the sum of the energies of each alone, because an interaction energy is $E(AB) - E(A) - E(B)$ and the method must give zero when the molecules do not interact. For two distant molecules A and B the exact wave function contains "a double on A and a double on B at once", which from the pair is a quadruple excitation that CISD does not contain, so correlation in A is squeezed out by correlation in B. The test with two

non-interacting H_2 molecules shows that the Hartree–Fock energy of the pair is exactly twice that of one, and MP2, CCSD and FCI are additive, but CISD is not.

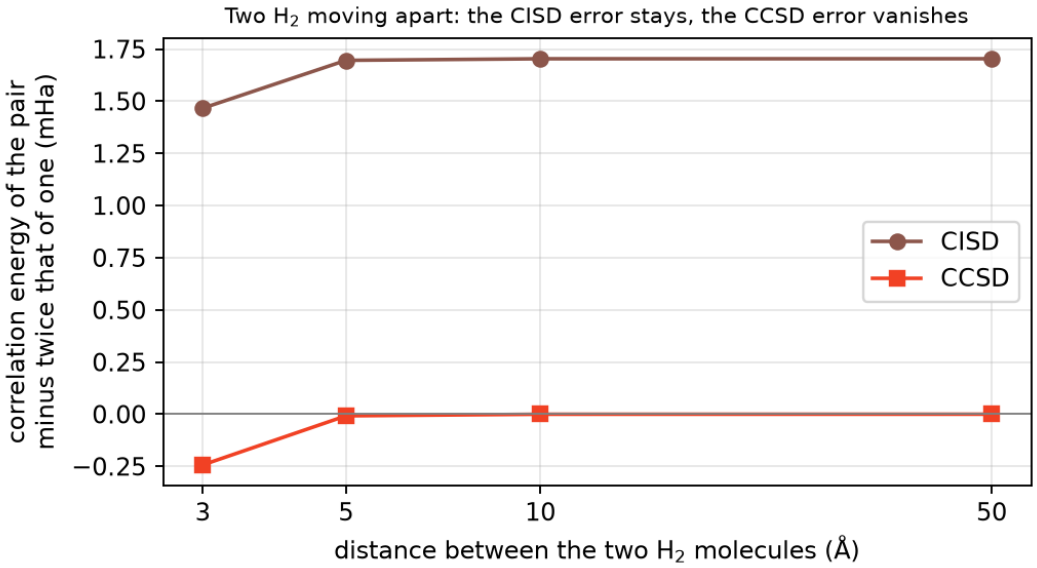


Figure 22: Two H_2 molecules at growing separation: correlation energy of the pair minus twice that of one. CCSD tends to zero; CISD levels off at a positive value.

10.3 *An exponential of excitations*

When A and B do not interact the wave function must be a product, $\Psi_{AB} = \Psi_A \Psi_B$, and the exponential is the function that turns sums into products. Let $a_p^\dagger$ put an electron into spin orbital p and a_p remove one. The operators for all single and double excitations, with amplitudes t_i^a and t_{ij}^{ab}, are

$$\hat{T}_1 = \sum_{ia} t_i^a \, a_a^\dagger a_i, \qquad \hat{T}_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \, a_a^\dagger a_b^\dagger a_j a_i, \qquad \hat{T} = \hat{T}_1 + \hat{T}_2 + \hat{T}_3 + \cdots,$$

and the coupled-cluster wave function is

$$\Psi_{CC} = e^{\hat{T}} \Phi_0 = \left(1 + \hat{T} + \tfrac{1}{2}\hat{T}^2 + \tfrac{1}{3!}\hat{T}^3 + \cdots\right)\Phi_0.$$

Keeping only $\hat{T} = \hat{T}_1 + \hat{T}_2$ is **CCSD**. Because the excitation operators commute, the exponential can be expanded like numbers: the doubles come from $\hat{T}_2 + \frac{1}{2}\hat{T}_1^2$, the triples from $\hat{T}_1\hat{T}_2 + \frac{1}{6}\hat{T}_1^3$, the quadruples from $\frac{1}{2}\hat{T}_2^2 + \cdots$. So CCSD has the same number of unknowns as CISD, yet its wave function reaches determinants with any number of excitations: the higher ones are *products* of the singles and doubles amplitudes (two pairs excited independently). What CCSD leaves out are the *connected* triples, quadruples and higher, which have their own amplitudes. And *size consistency is automatic*: for non-interacting A and B, $\hat{T} = \hat{T}_A + \hat{T}_B$ with commuting parts, so

$$e^{\hat{T}} \Phi_0 = \left(e^{\hat{T}_A} \Phi_A\right)\left(e^{\hat{T}_B} \Phi_B\right),$$

a product whose energy is the sum, and truncation at singles and doubles survives this. In the exact wave function of two independent H_2 molecules, the coefficient of the determinant with both molecules doubly excited equals the product of the two single-molecule double coefficients to six

digits: this is the $\frac{1}{2}\hat{T}_2^2$ term at work.

10.4 *The (T) correction*

The connected triples are left out of CCSD (the largest correction left at the usual single-reference level, from memory). Solving for them properly (CCSDT) is much more expensive, so the standard compromise is **CCSD(T)**: solve CCSD, then estimate the connected triples by perturbation theory once, from the converged amplitudes. In words, a triple excitation is reached from a double by exciting one more electron through the integrals; the energy is the square of that amplitude divided by an orbital-energy denominator, plus a cross term with the disconnected part built from the singles. The correction is computed once at the end, not iterated. CCSD costs about N^6 and CCSD(T) about N^7 in the size N of the system (from memory), against N^3 to N^4 for DFT, which is why the method is used as the reference for small systems and not for production.

IN FOUR LINES. (1) Hartree–Fock misses the correlation energy, and the exact answer needs every excitation from Φ_0 at a cost that grows exponentially. (2) Truncating the excitations (CISD) is not size consistent; writing the wave function as $e^{\hat{T}}\Phi_0$ with $\hat{T} = \hat{T}_1 + \hat{T}_2$ is. (3) The amplitudes follow from projecting $e^{-\hat{T}}\hat{H}e^{\hat{T}}\Phi_0 = E\Phi_0$, and the series ends at fourth order (from memory). (4) CCSD(T) is the reference for the dimers above, where dispersion is a correlation effect that Hartree–Fock cannot give.

11 TECHNICAL NOTES

11.1 *Orbitals and basis functions are different things*

In $\psi_i = \sum_\mu C_{\mu i}\phi_\mu$ the basis functions ϕ_μ are fixed ingredients and each orbital is a recipe that mixes them; the coefficients $C_{\mu i}$ connect the two sets of indices. The orbitals are orthonormal, $\int \psi_i \psi_j \, d\mathbf{r} = \delta_{ij}$, which in matrix form reads $\mathbf{C}^\mathsf{T}\mathbf{S}\mathbf{C} = \mathbf{1}$, while the basis functions overlap, so $\mathbf{S} \neq \mathbf{1}$. If the ϕ_μ were the true orbitals, $\mathbf{S}$ would be the identity and $\mathbf{F}$ diagonal, and there would be nothing to solve; the two meet in $\mathbf{C}^\mathsf{T}\mathbf{F}\mathbf{C} = \boldsymbol{\varepsilon}$. The program's labels for basis functions (`O 1s`, `O 2px`) say only what shape a function has: the oxygen core orbital is not simply the basis function called `O 1s` but a mixture of several s-type functions of different spread.

11.2 *Why the hydrogen ℓ-degeneracy is special*

The energy $-1/2n^2$ depends on n only because the potential is exactly $-1/r$. In a many-electron atom the potential seen by one electron is $-Z_{\text{eff}}(r)/r$ with Z_{eff} decreasing outward (section 5), which splits s, p and d; and for the oscillator or the drumhead the ladder is different again. The shapes (the angular parts) survive in every spherically symmetric problem; the degeneracy in ℓ does not.

11.3 *What was checked, and what was not*

Every number below was computed in the notebooks of the `qm-theory` repository, whose test suites run as `pytest` checks.

CHECKED NUMERICALLY.

- *Boxes and hydrogen:* the box energies and node counts, the degeneracies of the square and cube, the orthonormality of the spherical harmonics and of whole orbitals, $\ell(\ell+1)$ by finite

differences, the radial energies against $-1/2n^2$, the degeneracy n^2, and that the nodes of orbitals lie on the stated spheres, cones and planes.

- *Bridge:* the H_2^+ curve (minimum at $R \approx 2.0$ bohr; harmonic wavenumber 2312 cm^{-1} and level spacing 2189 cm^{-1} from the nuclear problem), the helium minimum $\zeta = 1.6875$ with $E = -2.84766$ and the electron-repulsion term $\frac{5}{8}\zeta$ by numerical integration, $s < p < d$ in the screened toy, the hybrid angles and orthonormality (also by integration over the sphere), the methane toy levels against its 2×2 blocks, and that the four bond orbitals span the occupied space exactly.
- *Basis sets:* the H_2^+ two-function integrals against quadrature, the STO-3G and STO-6G coefficients against PySCF, the Gaussian product theorem, the shell sizes and the water basis-function counts.
- *DFT:* the water SCF and the ωB97M-V exchange fractions (from PySCF; the pieces of F_{11} come from a one-off PySCF check that is not stored with the data), and the methane and water dimer interaction energies (stored PySCF runs).
- *Coupled cluster:* a numpy implementation of CISD, CCSD and (T) agrees with PySCF to better than 10^{-8} hartree on H_2 and water; the size-consistency test on two H_2 molecules.

NOT CHECKED HERE, OR STATED FROM MEMORY.

- The helium exact and Hartree–Fock-limit energies (-2.9037, -2.8617 hartree) and the real methane numbers (orbital energies, the 20/76/4 composition) are literature or stored values, not recomputed here.
- The cost exponents of HF, DFT, CCSD and CCSD(T), that the series of equations ends at fourth order, that the connected triples are the largest correction left after CCSD, and that CCSD(T) reaches a fraction of a kcal/mol for interaction energies in large basis sets are from memory.
- The toy models (screening, methane) use invented parameters and show only the ordering and the pattern of levels.
- The CCSD(T) reference for the dimers is in one basis (`def2-TZVPD`) and is not extrapolated to the complete-basis limit.
- The functional and basis-set citations (Becke 97, ωB97M-V, VV10, D3, `def2`) have not yet been verified against the papers.